14. option(c)

$$k = \left(\frac{1}{t}\right)\ln\left(\frac{[A_0]}{[A]}\right)$$

$[A_0]=0.1$; $[A]=0.05$

$$k = \left(\frac{1}{t_{1/2}}\right)\ln\left(\frac{0.1}{0.05}\right)$$

$$k = \left(\frac{1}{t_{1/2}}\right)\ln(2)$$

$$t_{1/2} = \frac{\ln(2)}{k}$$

15. option(b)

$$\text{rate}_1 = k[0.1]^n[0.1]^m \text{ ———(1)}$$

$$\text{rate}_2 = k[0.2]^n[0.1]^m \text{ ———(2)}$$

$(2)/(1)$

$$\frac{2x}{x} = \frac{k[0.2]^n[0.1]^m}{k[0.1]^n[0.1]^m}$$

$$\frac{2x}{x} = 2^n \quad \therefore\ n = 1$$

$$\text{rate}_3 = k[0.1]^n[0.2]^m \text{ ———(3)}$$

$$\text{rate}_4 = k[0.2]^n[0.2]^m \text{ ———(4)}$$

$(4)/(2)$

$$\frac{8x}{2x} = \frac{k[0.2]^n[0.2]^m}{k[0.2]^n[0.1]^m}$$

$$\frac{8}{2} = 2^m \quad \therefore\ m = 2$$

$$\therefore\ \text{rate} = k[A]^1[B]^2$$

16. option(c)

For a first reaction, If the concentration of reactant is doubled, then the rate of reaction also doubled.

Rate constant is independent of concentration and is a constant at a constant temperature,

17. option(a)

The unit of rate constant is s^{-1} and it indicates that the reaction is first order.

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